Thoughts And Rambles On Probability And Statistics

Problem Set 3: Expectation

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These are the end of chapter problems for Chapter 3 of the notes. Equation numbers refer to the full notes.

1. Let X and Y be independent random variables. Show that

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$$

2. Look back to the random sums example. Show that

$$\text{Var}[S_N] = \text{Var}[X_1]\,\mathbb{E}[N] + \mathbb{E}[X_1]^2\,\text{Var}[N]$$

by first finding the second moment and then subtracting the square of the expectation.

3. (Computational: Simulating the Stock Market.) Let $Y_1, Y_2, \ldots$ be independent random variables such that $\mathbb{P}(Y_i = 1) = \mathbb{P}(Y_i = -1) = \frac{1}{2}$. Let $X_n = \sum_{i=1}^{n} Y_i$. Think of $Y_i = 1$ as "the stock price increased by one dollar," $Y_i = -1$ as "the stock price decreased by one dollar," and X_n as the value of the stock on day n.

(a) Find $\mathbb{E}(X_n)$ and $\text{Var}(X_n)$.

(b) Simulate X_n and plot X_n versus n for $n = 1, 2, \ldots, 10,000$. Repeat the whole simulation several times. Notice two things. First, it's easy to "see" patterns in the sequence even though it is random. Second, you will find that the four runs look very different even though they were generated the same way. How do the calculations in (a) explain the second observation?

4. Let X be a continuous random variable with cdf F. Suppose that $\mathbb{P}(X > 0) = 1$ and that $\mathbb{E}(X)$ exists. Show that:

$$\mathbb{E}(X) = \int_0^\infty \mathbb{P}(X > x)\,dx.$$

Hint: Consider integrating by parts. The following fact is helpful: if $\mathbb{E}(X)$ exists, then $\lim_{x\to\infty} x[1 - F(x)] = 0$.

5. Let X be a non negative random variable. Show that

$$\lim_{n\to\infty} \frac{1}{n}\mathbb{E}\left(\frac{1}{X}\mathbf{1}\{X > \frac{1}{n}\}\right) = 0.$$

6. Let $\{A_n, n \geq 1\}$ be independent events, and suppose that $P(A_n) < 1$ for all n. Prove that

$$P(A_n \text{ i.o. }) = 1 \Longleftrightarrow P\left(\bigcup_{n=1}^{\infty} A_n\right) = 1.$$

Why is $P(A_n) = 1$ forbidden?

7. Let $X \sim N(0,1)$ and let $Y = e^X$. Find $\mathbb{E}(Y)$ and $\mathrm{Var}(Y)$.